UNIVERSITY OF ILLINOIS CHICAGO

DEPARTMENT OF CHEMICAL ENGINEERING

TRANSPORT PHENOMENA I

FALL 2025

Department of Chemical Engineering
College of Engineering
929 W. Taylor St., 226 EIB (MC 110)
Chicago, IL 60607

Phone 312.996.3424
Fax 312.996.0808
Email cheuic@uic.edu
Web che.uic.edu

COLLEGE OF ENGINEERING, UIC

TRANSPORT PHENOMENA I

CHE 311, CREDIT HOURS: 3
FALL 2025

I. INSTRUCTOR & COURSE DETAILS

INSTRUCTOR

KOROSH TORABI
Email address: korosh@uic.edu
Drop-In Office Hours: Tuesdays 11 AM – 1 PM, EIB 246

TEACHING ASSISTANT

KIM NGO
E-mail: bngo@uic.edu
Drop-In Office Hours: Tuesdays 1:00 PM - 2:00 PM and Thursdays 2:00 PM - 4:00 PM, EIB 247

DI XIAO
E-mail: dxiao9@uic.edu

COURSE BLACKBOARD SITE

https://uic.blackboard.com
Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. Email the Learning Technology Solutions team at LTS@uic.edu for all technical questions about Blackboard.

COURSE MODALITY AND SCHEDULE

This course is taught on campus.
Tuesdays and Thursdays 9:30 AM - 10:45 AM
EIB Room 124

II. COURSE INFORMATION

CATALOG COURSE DESCRIPTION AND PREREQUISITE/COREQUISITE STATEMENT

Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies. Course Information: Prerequisite(s): Credit or concurrent registration in CHE 210; and MATH 220; and CHE 205.

COURSE RATIONALE AND GROWTH MINDSET

This course presents the main physical ideas concerning fluid dynamics and momentum transport. Knowledge of these topics is needed in subsequent undergraduate chemical engineering courses, specifically heat transfer (energy transport), and mass diffusion (mass transport). In a sense, this is a transition course between elementary physics and chemical engineering; the topics covered are subjects in classical physics, but the applications given are directed toward chemical engineers.

It is assumed that the students in the course have a thorough grounding in elementary mathematics, and this mathematical background will be used; some mathematics review may be necessary at various points in the course.

It is strongly suggested that you read the textbook carefully, do the homework problems in detail without consulting other solutions, ask thoughtful questions in class and during office hours, discuss the material with your classmates, and work on additional problems to master it.

Course materials and assignments can be complex and challenging, but they are crucial to intellectual and personal growth and development. There are times when you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, visit the tutoring center, and contact faculty when struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

COURSES LEARNING OUTCOMES

At the completion of this course, students should be able to:

- Use mathematical language to calculate scalar, vector, and tensor operations relevant to transport phenomena.
- Understand the essential physical principles of momentum transport and use the conservation statement for momentum to solve elementary fluid dynamics problems.
- Set up differential momentum balance and analyze boundary conditions.
- Solve ordinary differential equations and apply boundary conditions to determine steady-state one-dimensional flow velocity and stress profiles.
- Solve partial differential equations by using the methods of separation of variables and combination of variables that result in the velocity profiles of the flow fields, which have two independent variables.
- Apply time-smoothed equations of change for turbulent flow.
- Understand the physical meaning and mathematical expression of the empirical expressions for the turbulent momentum flux.
- Understand the definition of friction factors to solve various pipe-design problems.
- Set up macroscopic mass, momentum, and mechanical energy balances for isothermal flow systems.

ABET STUDENT LEARNING OUTCOMES CENTRAL TO THE COURSE

- ABET Student Outcome 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- ABET Student Outcome 2: an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

- ABET Student Outcome 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

BRIEF LIST OF TOPICS TO BE COVERED

Mathematics review (Vector and Tensor Algebra and calculus), momentum flux, newton's law of viscosity, shell balances (unidirectional flow problems), equations of change, two-dimensional flow, turbulent flow, friction factors and macroscopic momentum balance.

COURSE MATERIALS

REQUIRED TEXTBOOK

Transport Phenomena 2nd Edition
by R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot
Note: revised 2nd edition is preferred.

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

GRADING POLICY AND POINT BREAKDOWN

- Homework: 20%
- Quizzes: 10%
- Midterm Exam I: 20%
- Midterm Exam II: 20%
- Final Exam 30%

ASSIGNMENTS

- Reading assignments: Assigned from the textbook. No submission is required.
- Problem sets (Homework): Please submit an electronic copy of your homework by the specified due date before the start of the class.

POLICY FOR MISSED OR LATE WORK

- Homework and Term Project: Late submissions will not be considered for partial credit. Unreadable and unorganized homework will not be graded.
- Quizzes: No makeup will be provided for missed quizzes; however, the lowest quiz grade will be dropped from the final grade calculation.
- Exams: Students who must miss an exam for any reason are expected to contact the instructor before the exam date.

ATTENDANCE AND PARTICIPATION POLICY

Please email me if you face an unexpected situation that may impede your attendance and participation in required class and exam sessions.

EMAIL EXPECTATIONS

Students are responsible for all information instructors send to your UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

ACADEMIC INTEGRITY

As a student and member of the UIC community, you are expected to adhere to the Community Standards of academic integrity, accountability, and respect. Please review the UIC Student Disciplinary Policy for additional information.

OTHER COURSE POLICIES

Regrading of exams must be requested within 2 days after receiving the grades. For any homework issues, please reach out to the TA.

IV. COURSE SCHEDULE

Week	Topics	Book Chapter
1	Conservation laws, flux and balance equations Math Review	Chapter 0, Appendix A and Math Handout
2	Continuum fluid mechanics and Stress tensor Math review	Appendix A and Math Handout
3	Newton's law of viscosity and the combined momentum flux	Chapter 1
4	Introduction to shell momentum balance Flow of a falling film	Chapter 2
5	Flow through a circular tube and modified pressure Creeping flow around a sphere	Chapter 2
6	**Midterm Exam I**	Appendix A and Math Handout Chapters 0, 1 and 2
6	The equation of continuity and the equations of motion Drifting time derivative and Reynolds transport theorem	Chapter 3
7	Solving flow problems using the equations of change	Chapter 3
8	The equation of mechanical energy and viscous dissipation	Chapter 3
9	Unsteady flow and method of combination of variables Unsteady flow and method of separation of variables	Chapter 4
10	**Midterm Exam II**	Chapters 3 and 4
10	Introduction to turbulent flow and time-smoothed equations of change Empirical equations for momentum flux and eddy viscosity	Chapter 5
11	March 24-28. Spring break. No classes.	
12	Friction factors for flow in tubes	Chapter 6
13	Friction factors for flow around submerged objects Pressure drop in packed beds	Chapter 6
14	Macroscopic momentum balance Macroscopic mechanical energy balance, the Bernoulli equation and estimation of the viscous loss	Chapter 7

15	Solving flow problems using the macroscopic balances	Chapter 7
16	Review and problem solving	
17	**Final Examination**	Comprehensive

- Quizzes will be scheduled during regular class time and location with at least two days' notice.
- Midterm exams will take place at regular class time and location.
- The final exam will be held at the date and time specified by the university's Registrar's Office.

DISCLAIMER

This syllabus is intended to give the student guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance through in-class announcements and in writing via Blackboard Announcements.

V. ACCOMMODATIONS

DISABILITY ACCOMMODATION PROCEDURES

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc@uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodation, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

RELIGIOUS ACCOMMODATIONS

Following campus policy, if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit this form by email with the subject heading: “YOUR NAME: Requesting Religious Accommodation.”

VI. CLASSROOM ENVIRONMENT

INCLUSIVE COMMUNITY

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

NAME AND PRONOUN USE

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are he/him. I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: https://www.mypronouns.org/what-and-why.

COMMUNITY AGREEMENT/CLASSROOM CONDUCT POLICY

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another's privacy, and do not invite outsiders into our classroom.

CONTENT NOTICES AND TRIGGER WARNINGS

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

VII. RESOURCES: ACADEMIC SUCCESS, WELLNESS, AND SAFETY

We all need the help and the support of our UIC community. Please visit my office hours for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

ACADEMIC SUCCESS

- UIC Tutoring Resources
- College of Engineering tutoring program
- Equity and Inclusion in Engineering Program
- UIC Library and UIC Library Research Guides
- Offices supporting the UIC Undergraduate Experience and Academic Programs.
- Student Guide for Information Technology

WELLNESS

- Counseling Services: You may seek free and confidential services from the Counseling Center at https://counseling.uic.edu/.
- Access U&I Care Program for assistance with personal hardships.

- Campus Advocacy Network: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at http://can.uic.edu/ .

SAFETY

- UIC Safe App—PLEASE DOWNLOAD FOR YOUR SAFETY!
- UIC Safety Tips and Resources
- Night Ride
- Emergency Communications: By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.